ModelIt!

Student Activity Pack

De-Extinction: Should We Bring Back the Woolly Mammoth?

Modeling the Genetics, Ecology, and Ethics of Resurrecting Extinct Species

Grade 10L3 | NGSS HS-LS4-4, HS-LS2-7 | Stability and Change

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. **Do you think scientists should try to bring back extinct animals like the woolly mammoth? Why or why not?**

2. **What challenges do you think would arise from placing an extinct animal in today's world?**

3. **Draw what you think the process of de-extinction would involve, from ancient DNA to living animal.**

4. **If you had $150 million for conservation, would you spend it on de-extinction or protecting currently endangered species? Explain your reasoning.**

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

DNA Recovery Quality | Gene Editing Accuracy | Surrogate Compatibility | Population Viability | Ecosystem Readiness | Habitat Availability | Genetic Diversity | Disease Resistance | Ecological Integration | Ethical Threshold

EXTERNAL	INTERNAL

? Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Best Case Technology

Highest achievable DNA quality and gene editing precision

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Ecological Mismatch

Viable organism but realistic modern habitat conditions

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Ethics vs. Science

Technically viable but high ethical scrutiny

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Behavioral Repertoire — The set of learned behaviors — migration routes, social structures, foraging strategies, predator avoidance — that the de-extinct species would need but cannot inherit genetically, as they were culturally transmitted between generations of the original species

Climate Trajectory — The projected climate conditions in potential habitat areas over the next 50-100 years, which may continue shifting away from the conditions the species originally evolved to inhabit

Keystone Effects — The potential ecosystem engineering benefits of mammoth reintroduction — such as maintaining grasslands through grazing, compacting snow to increase permafrost preservation, and potentially slowing Arctic carbon release

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

If we have the technology to bring back extinct species, does that mean we should — and could they even survive in a world that has moved on without them? Use evidence from your simulation to support your answers.

1 How does your model demonstrate that de-extinction is not just a genetics problem but an integrated ecological challenge?

2 What does the minimum viable population requirement reveal about the difference between creating one animal and restoring a species?

3 How does the model treat Ethical Threshold as a variable that interacts with scientific feasibility?

4 If mammoth reintroduction could slow permafrost thaw and reduce carbon emissions, does that change the ethical calculus?

5 What are the limitations of modeling a de-extinction project that has never been attempted at this scale?

8 STEM Challenge: Design a De-Extinction Viability Assessment

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YOUR ENGINEERING MISSION

Develop a comprehensive viability assessment for a woolly mammoth de-extinction project that evaluates all ten variables and recommends whether the project should proceed, be modified, or be redirected toward existing conservation efforts.

THE SCENARIO

A biotech company has secured $150 million in funding for a woolly mammoth de-extinction project. Before the project proceeds, they need an independent viability assessment from a team of young scientists. Your team must evaluate every stage of the pipeline — from DNA recovery through ecosystem integration — and deliver a recommendation with evidence from your computational model.

Engineering Design Thinking

1. WHAT IS YOUR ASSESSMENT OF EACH OF THE TEN MODEL VARIABLES — IS EACH CURRENTLY AT A VIABLE LEVEL?
2. WHICH VARIABLES REPRESENT THE BIGGEST BARRIERS AND WHAT WOULD BE NEEDED TO OVERCOME THEM?
3. WHAT IS THE MINIMUM TIMELINE FROM FIRST GENE-EDITED CELL TO SELF-SUSTAINING WILD POPULATION?
4. WHAT IS YOUR TEAM'S RECOMMENDATION: PROCEED, MODIFY, OR REDIRECT — AND WHAT EVIDENCE FROM YOUR MODEL SUPPORTS IT?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Conservation Geneticists use genomic tools to protect endangered species, manage genetic diversity in captive breeding programs, and assess the viability of small populations. They work at zoos, wildlife agencies, universities, and organizations like Revive & Restore, earning $60,000-$120,000/year. Bioethicists who evaluate the ethical dimensions of biotechnology applications earn $60,000-$130,000/year.